Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.08869109910324896, median 0.08161567516364657, std: 0.055566507242716494
Reprojection error (cam1): mean 0.08694859515454166, median 0.08058301070121461, std: 0.057502935292181596
Gyroscope error (imu0): mean 0.31171529978303103, median 0.09205573680401676, std: 0.4458600099750376
Accelerometer error (imu0): mean 0.3488732502152947, median 0.2326210766989035, std: 0.2679300756029612

Residuals

Reprojection error (cam0) [px]: mean 0.08869109910324896, median 0.08161567516364657, std: 0.055566507242716494
Reprojection error (cam1) [px]: mean 0.08694859515454166, median 0.08058301070121461, std: 0.057502935292181596
Gyroscope error (imu0) [rad/s]: mean 0.054527907537664067, median 0.01610317718846285, std: 0.07799364809999068
Accelerometer error (imu0) [m/s^2]: mean 0.4068526278807424, median 0.2712804615916851, std: 0.3124574764046263

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.54852445 -0.81259827 -0.19698979 0.17460212]
 [0.28666424 0.40408188 -0.86864345 0.09723334]
 [0.78545817 0.42000224 0.45459166 -0.13074288]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.54852445 0.28666424 0.78545817 -0.02095379]
 [-0.81259827 0.40408188 0.42000224 0.15750345]
 [-0.19698979 -0.86864345 0.45459166 0.17829056]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.0

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.55018625 -0.81167014 -0.19618019 0.1243166]
[0.28700808 0.40443079 -0.86836749 0.09711339]
[0.78416927 0.42145856 0.45546815 -0.1310614]
[0.   0.   0.   1.   ]]

T_ic: (cam1 to imu0):
[[0.55018625 0.28700808 0.78416927 0.00650471]
[-0.81167014 0.40443079 0.42145856 0.11686538]
[-0.19618019 -0.86836749 0.45546815 0.16841286]
[0.   0.   0.   1.   ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.0

Baselines:

Baseline (cam0 to cam1):
[[0.99999786 0.00014816 0.00206313 -0.05002982]
[-0.00014928 0.99999984 0.00054206 -0.000023]
[-0.00206305 -0.00054237 0.99999772 0.00009413]
[0.   0.   0.   1.   ]]
baseline norm: 0.050029909621438384 [m]

Gravity vector in target coords: [m/s^2]
[2.88811176 -9.2901415   1.2330873]

Calibration configuration
=========================

cam0

Camera model: pinhole
Focal length: [387.2761661519647, 387.3360745966035]
Principal point: [322.55427593991755, 233.459327809873]
Distortion model: radtan
Distortion coefficients: [0.005370231457773946, -0.004090260028352018, -0.0014951575909389215, 0.0009066421916205414]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

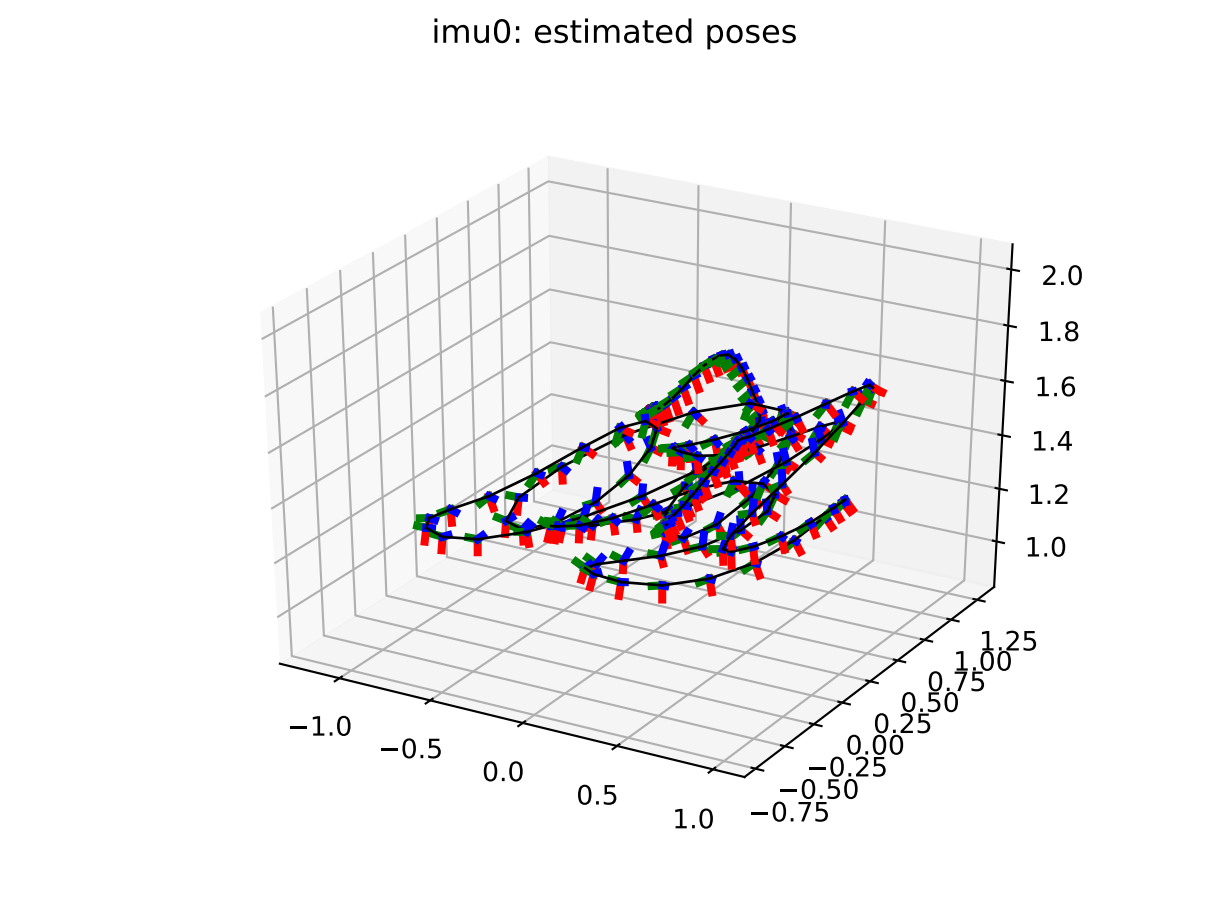
Camera model: pinhole
Focal length: [387.2715250757298, 387.35460697137654]
Principal point: [322.199824726234, 232.79121749202474]
Distortion model: radtan
Distortion coefficients: [0.006713647136759425, -0.005177384248077199, -0.0017880988996787479, 0.0005288146263442444]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

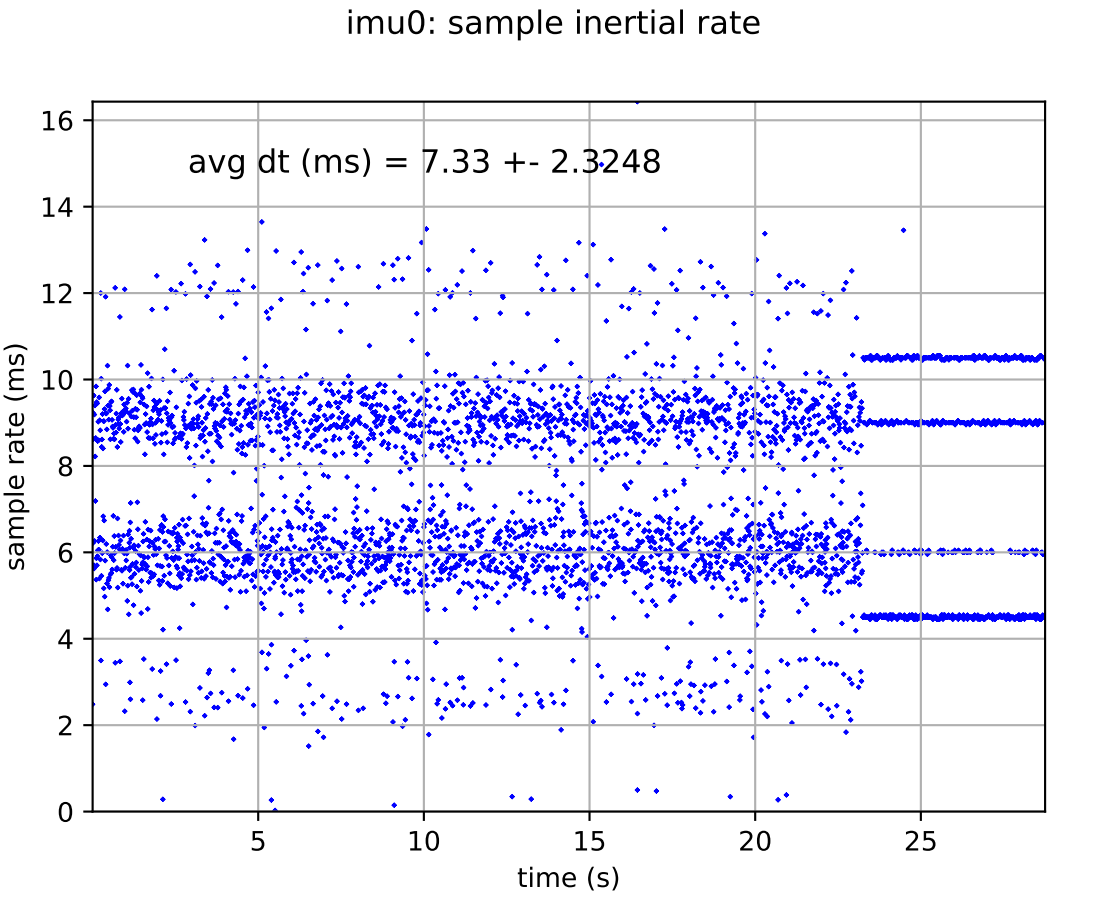
IMU configuration
==========

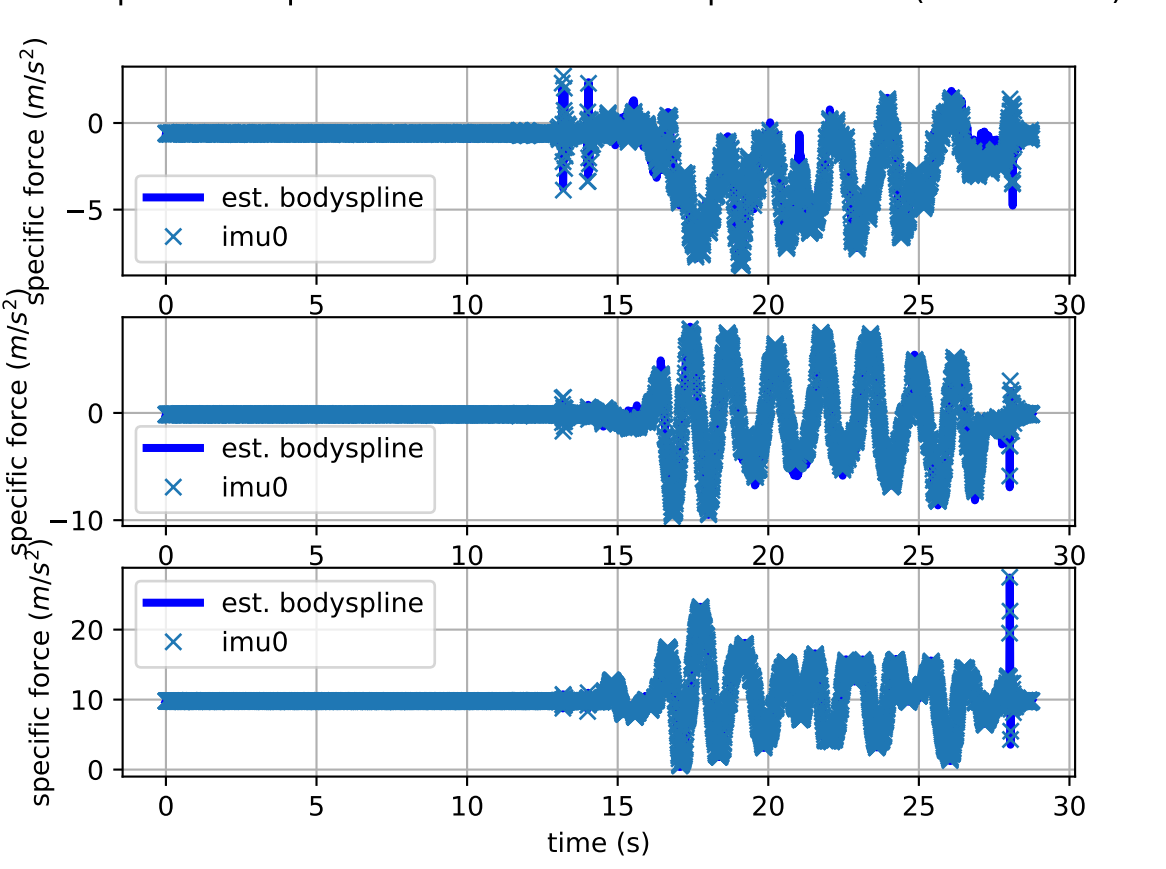
IMU0:

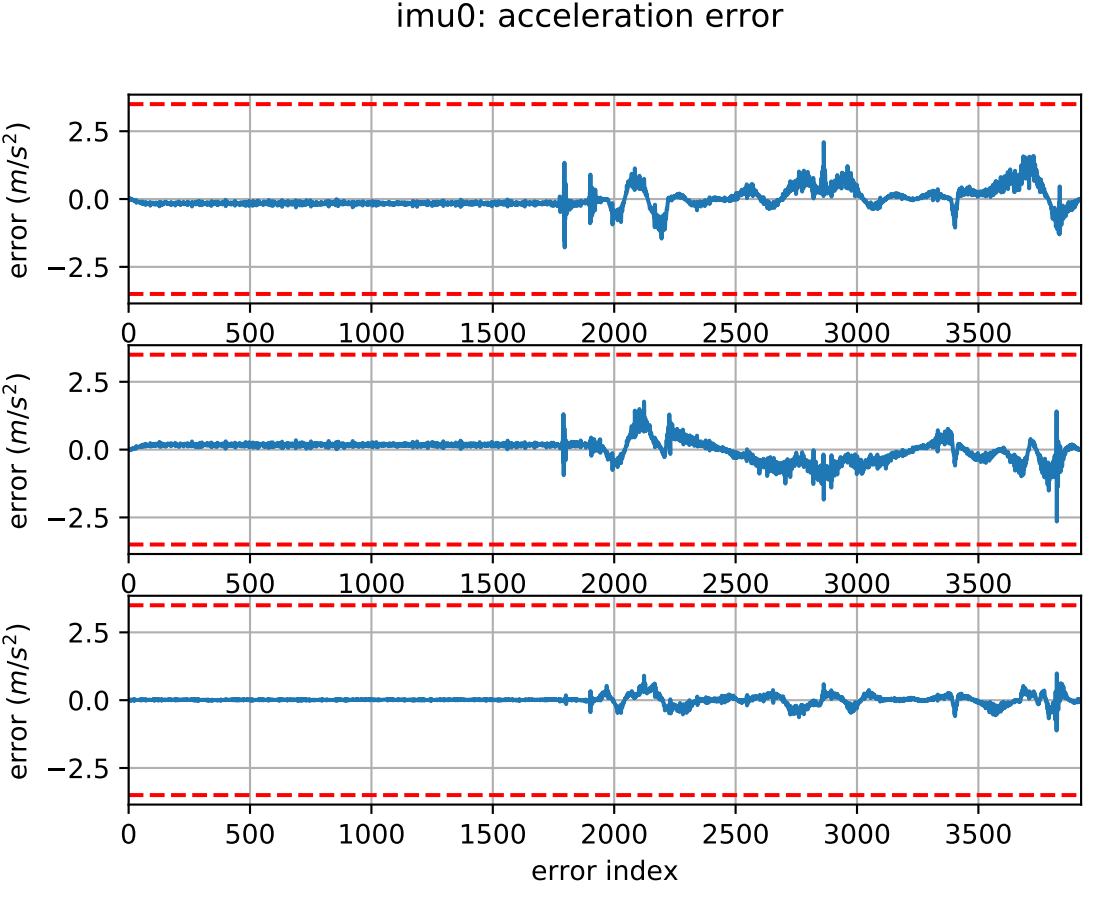
Model: calibrated
Update rate: 136

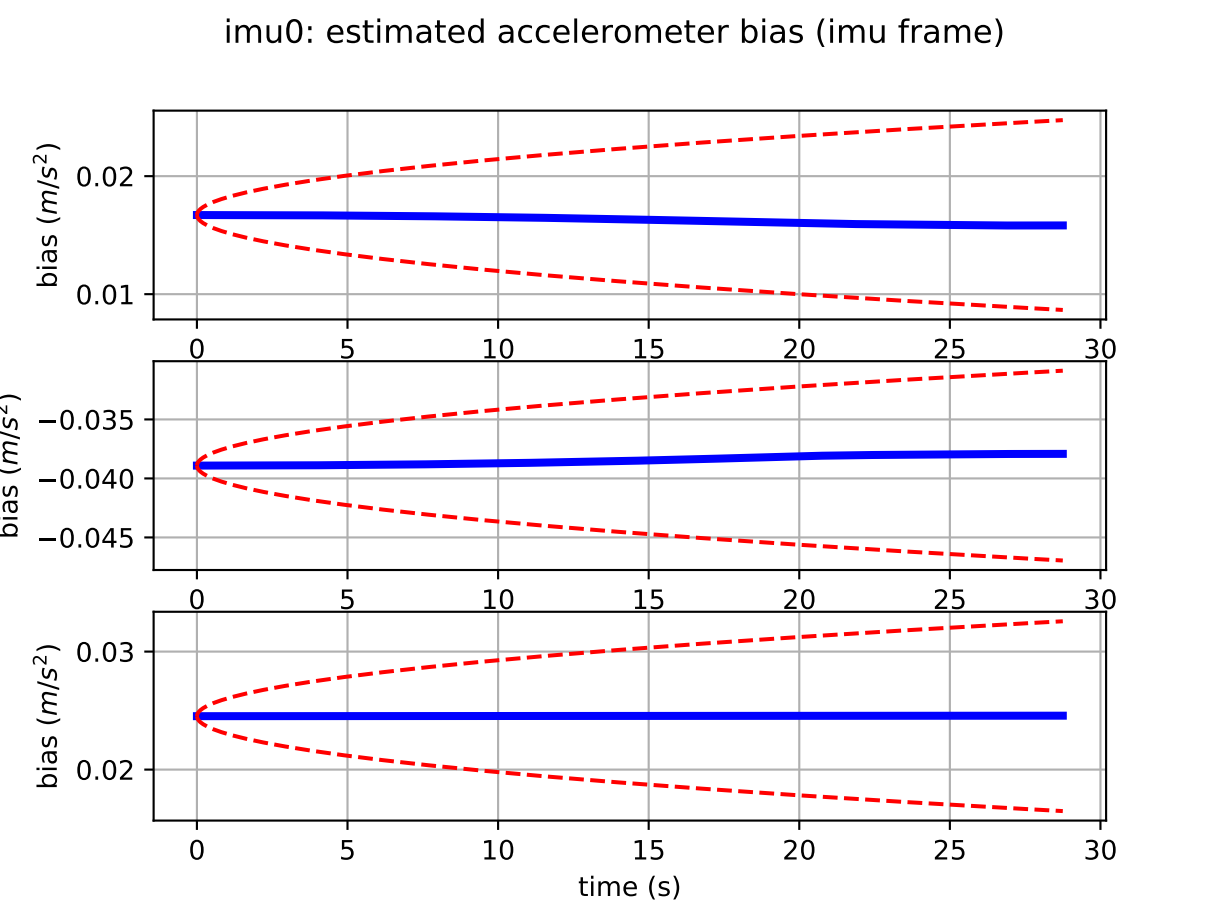
Accelerometer:
Noise density: 0.1
Noise density (discrete): 1.1661903789690602
Random walk: 0.0005
Gyroscope:
Noise density: 0.015
Noise density (discrete): 0.174928556845359
Random walk: 0.0002
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

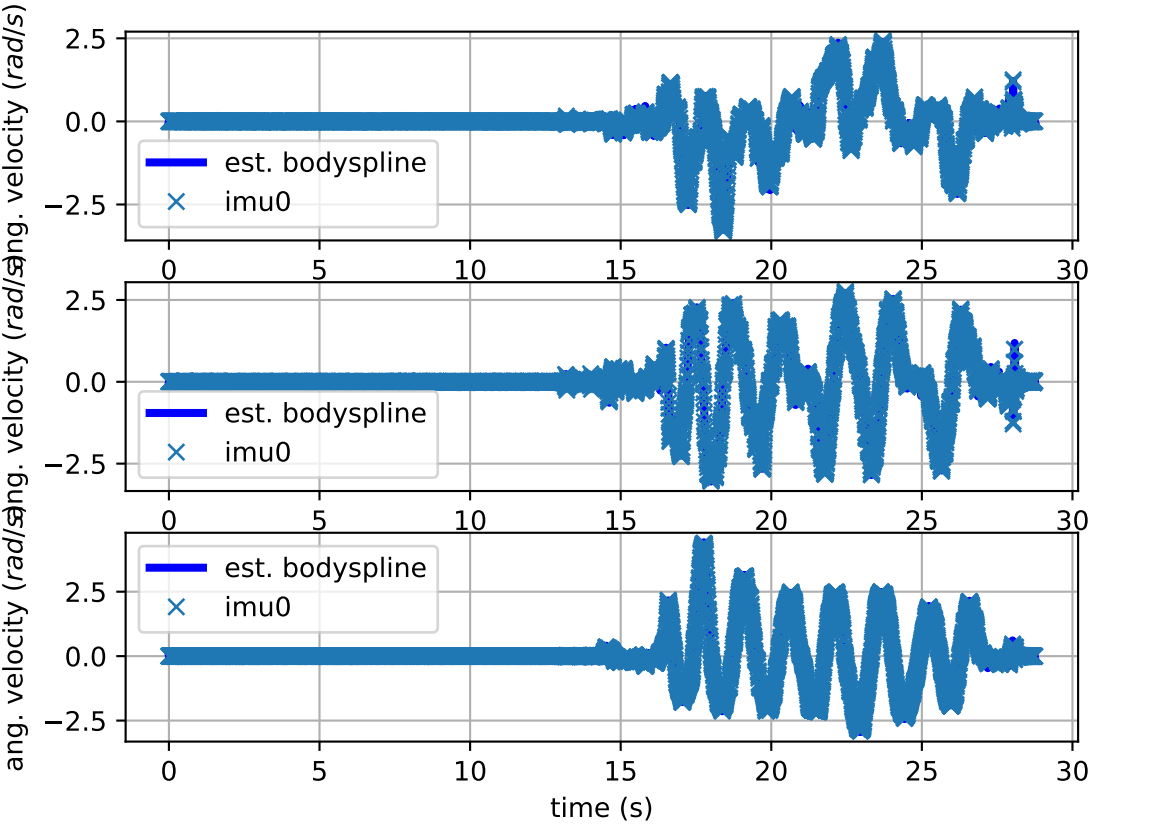

imu0: estimated poses
2.0
1.8
1.6
1.4
1.2
1.0
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.0
−0.5
0.0
0.5
1.0

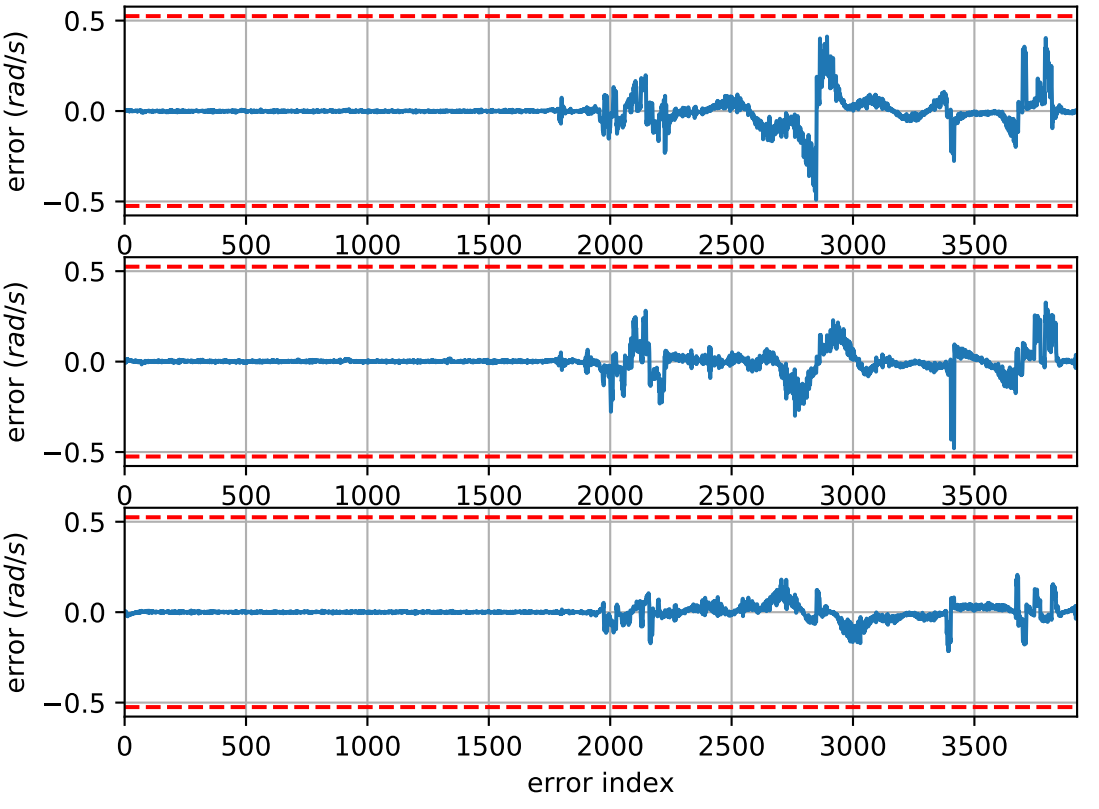

imu0: sample inertial rate
avg dt (ms) = 7.33 +- 2.3248
sample rate (ms)
time (s)
0
2
4
6
8
10
12
14
16
5
10
15
20
25

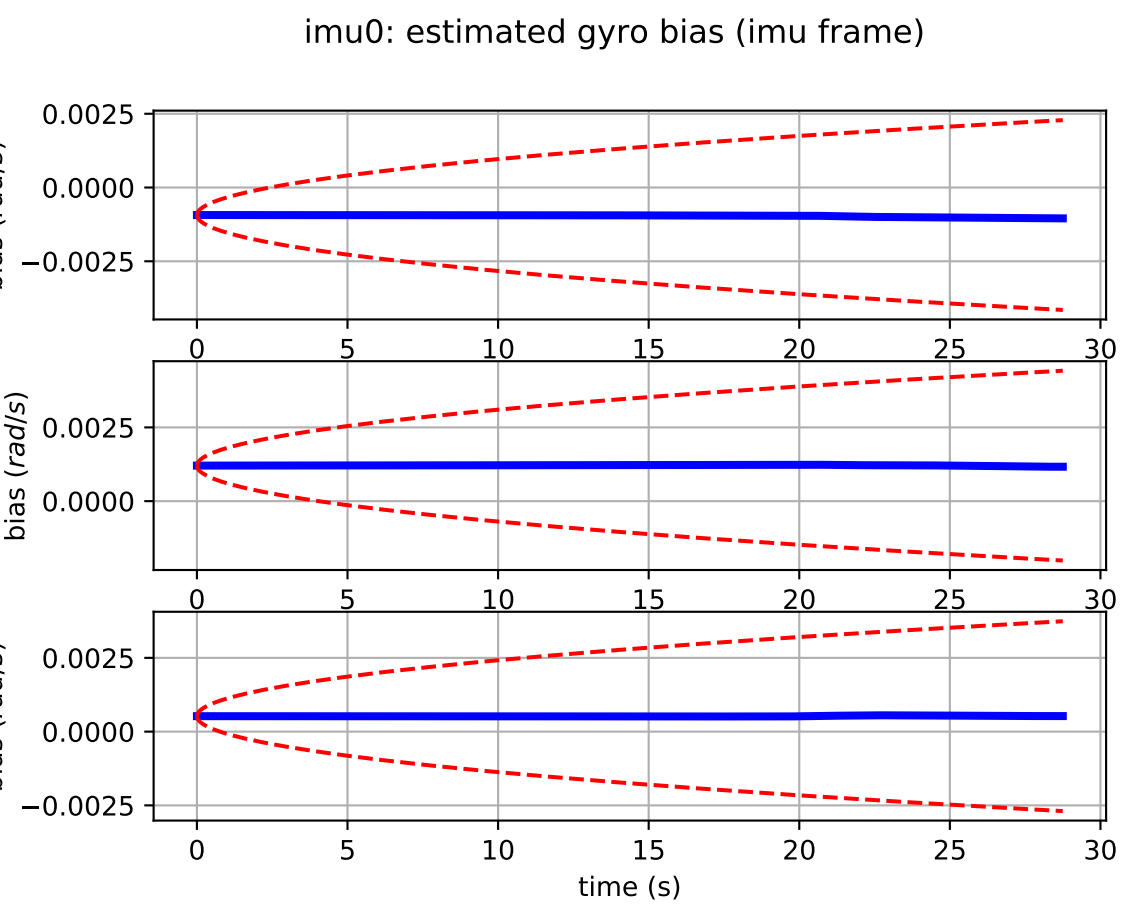

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
0
−5
0
5
10
15
20
25
30
specific force (m/s^2)
est. bodyspline
imu0
0
−10
0
5
10
15
20
25
30
specific force (m/s^2)
est. bodyspline
imu0
20
10
0
0
5
10
15
20
25
30
time (s)

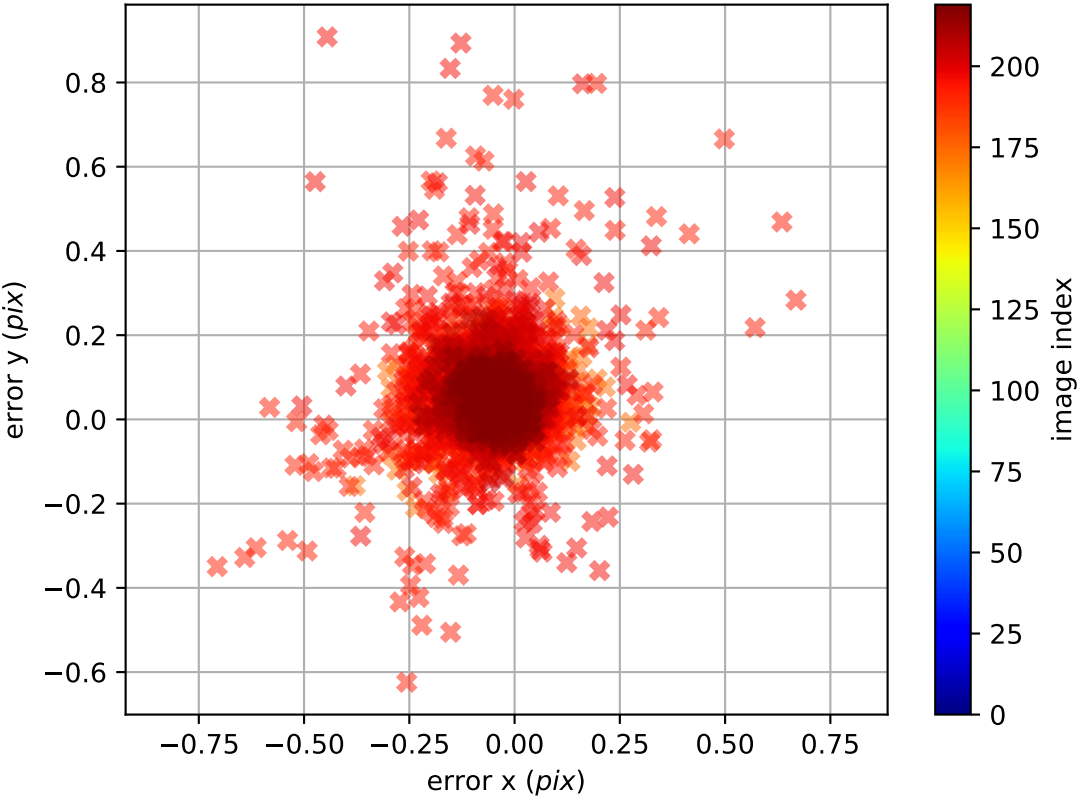

imu0: acceleration error
error (m/s^2)
2.5
0.0
−2.5
0
500
1000
1500
2000
2500
3000
3500
error index

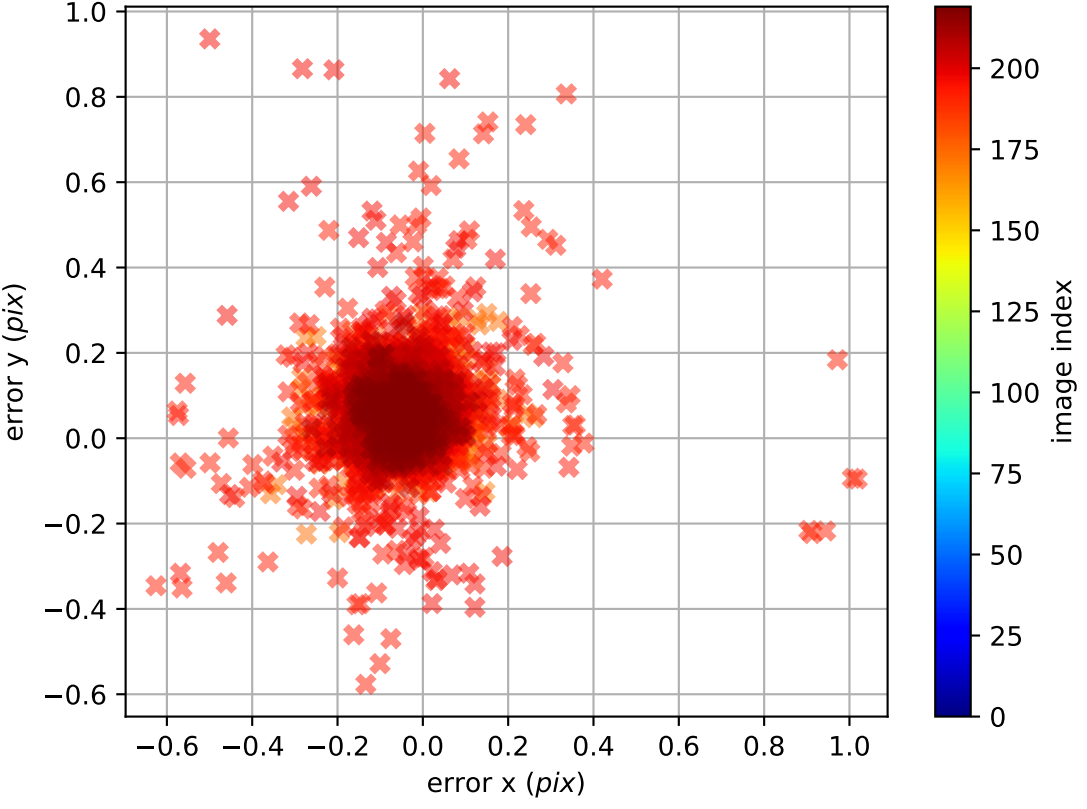

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.02
0.01
0
5
10
15
20
25
30
bias (m/s^2)
−0.035
−0.040
−0.045
0
5
10
15
20
25
30
bias (m/s^2)
0.03
0.02
0
5
10
15
20
25
30
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
2.5
0.0
−2.5
0
5
10
15
20
25
30
time (s)


imu0: angular velocities error
error (rad/s)
0.5
0.0
−0.5
0
500
1000
1500
2000
2500
3000
3500
error index


imu0: estimated gyro bias (imu frame)
0.0025
0.0000
−0.0025
0
5
10
15
20
25
30
bias (rad/s)
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
200
175
150
125
100
75
50
25
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
200
175
150
125
100
75
50
25
0